

Deep Learning

is a subset of Machine Learning that uses Artificial Neural Networks with multiple hidden layers to learn complex patterns from large amounts of data.

It is inspired by the working of the human brain and biological neurons. Deep Learning models can automatically learn features from data without requiring manual feature extraction.

The basic structure of a Deep Learning model consists of three main layers:

1. Input Layer
2. Hidden Layer(s)
3. Output Layer

A Neural Network: is a system of interconnected artificial neurons inspired by the human brain. It is mainly used for pattern recognition, prediction, classification, and regression tasks.

The working of a neural network follows these steps:

Step 1: Receive Input: The neuron receives input values.

Step 2: Multiply Inputs with Weights: Each input is multiplied with a weight., Weights represent the importance of inputs.

Step 3: Add Bias: Bias helps shift the activation function and improves learning.

Step 4: Apply Activation Function: Activation function introduces non-linearity.

Common activation functions: ReLU , Sigmoid , Tanh , Softmax

Step 5: Generate Output: The neuron generates prediction/output.

Step 6: Calculate Error: The predicted output is compared with actual output., Loss function calculates error.

Step 7: Backpropagation: Weights are updated to reduce error.

This process improves model accuracy.

The mathematical representation is:

$$Y = \Sigma(WX) + b$$

where:

- W represents weights
- X represents input values
- b represents bias

The activation function helps the network learn non-linear patterns.

There are mainly three types of Neural Networks:

1. ANN (Artificial Neural Network)
2. CNN (Convolutional Neural Network)
3. RNN (Recurrent Neural Network)

Difference Between ANN, CNN, and RNN

ANN	CNN	RNN
Works on structured data	Works on image data	Works on sequential data
No memory	Spatial feature extraction	Has memory
Fully connected layers	Convolution + Pooling	Recurrent connections
Used for prediction/classification	Used for computer vision	Used for NLP/time series

1. ANN (Artificial Neural Network)

Artificial Neural Network (ANN) is the basic type of neural network inspired by the human brain.

It consists of interconnected neurons arranged into:

1. Input Layer
2. Hidden Layer(s)
3. Output Layer

ANN is mainly used for:

- Classification
 - Regression
 - Prediction
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Working of ANN (Detailed)

Step 1: Input Layer: Receives input data.

Example: Student marks , Loan data

Step 2: Hidden Layer

Each neuron:

- Multiplies inputs with weights
- Adds bias
- Applies activation function

The hidden layer learns patterns from data.

Step 3: Output Layer

Produces final prediction.

Example:

- Loan Approved

Step 4: Error Calculation

Loss function measures prediction error.

Step 5: Backpropagation

Weights are updated using gradient descent.

Purpose:

- Reduce error
 - Improve accuracy
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Applications of ANN

- Student result prediction
 - House price prediction
 - Loan approval prediction
 - Diabetes prediction
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2. CNN (Convolutional Neural Network)

Convolutional Neural Network (CNN) is a Deep Learning architecture specially designed for image processing and computer vision tasks.

CNN automatically learns features like: Edges , Textures , Shapes

Why CNN was Introduced?

Traditional ANN struggles with images because:

- Images contain millions of pixels
- Too many parameters
- High computational complexity

CNN solves this using:

- Convolution

- Pooling
 - Feature extraction
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Architecture of CNN

Input Image → Convolution Layer → ReLU → Pooling → Flatten → Fully Connected Layer → Output

Working of CNN

Step 1: Input Image: CNN receives image input.

Types:

- Grayscale image → 1 channel
 - RGB image → 3 channels
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Step 2: Convolution Layer

Filters/kernels slide over image.

Purpose: Detect edges , Detect textures , Detect shapes

Feature maps are generated.

What is Filter/Kernel?

A small matrix (example: 3×3) that extracts features from image.

Feature Map

Output generated after convolution.

Represents detected features.

Step 3: Activation Function

ReLU introduces non-linearity.

$$f(x) = \max(0, x)$$

Step 4: Pooling Layer

Reduces feature map size.

Types:

1. Max Pooling
2. Average Pooling

Purpose:

- Reduce computation
- Reduce overfitting

Step 5: Flatten Layer

Converts 2D feature maps into 1D vector.

Step 6: Fully Connected Layer

Performs final classification.

Example:

- Cat
- Dog
- Tumor

Step 7: Output Layer

Generates final prediction.

Important CNN Terms

Stride: Controls movement of filter

- Stride = 1 → detailed/slow extraction
 - Stride = 2 → faster computation
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Padding: Adds extra pixels around image borders.

Purpose:

- Preserve image size
 - Prevent information loss
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Applications of CNN

- Face recognition
 - Self-driving cars
 - Medical image analysis
 - Image classification
 - Object detection
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Popular CNN Models

- LeNet
- AlexNet
- VGG16/VGG19
- ResNet
- MobileNet
- Inception

3. RNN (Recurrent Neural Network):

Recurrent Neural Network (RNN) is a neural network designed for sequential and time-series data.

RNN has memory capability, meaning it remembers previous information to predict future outputs.

Main Feature of RNN

RNN stores previous outputs and uses them for future predictions.

This makes RNN suitable for sequential data.

Working of RNN:

Step 1: Receive Current Input

RNN receives current input data.

Example: Word in sentence

Step 2: Combine with Previous Hidden State

RNN combines:

- Current input
- Previous memory

This helps understand sequence dependency.

Step 3: Process Information

The hidden layer processes current + previous information.

Step 4: Generate Output

Produces prediction.

Example:

- Next word prediction
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Step 5: Pass Memory Forward

The hidden state is passed to next time step.

This creates memory capability.

Applications of RNN

- Language translation
 - Speech recognition
 - Text generation
 - Sentiment analysis
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Problems in RNN

Vanishing Gradient Problem

During backpropagation:

- Gradients become very small
 - Model forgets long-term dependencies
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Solution: LSTM

LSTM (Long Short-Term Memory):

- Special type of RNN
 - Remembers long-term information
 - Solves vanishing gradient problem
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Best One-Line Definitions

ANN

ANN is a basic neural network inspired by the human brain used for prediction, classification, and regression tasks.

CNN

CNN is a Deep Learning architecture specially designed for image processing and computer vision tasks.

RNN

RNN is a neural network designed for sequential data that uses memory to remember previous information.

